2027 MOU reallocation, dollar for dollar, HIV money only: 15 countries (no model changes)

Generated by `audit/reallocation/run_dollar_hiv_2027.py` from `allocations_2027_v2.csv`. Every dollar moved between countries is treated as HIV money: US$1 of MOU = US$1 of HIV aid, spent the way the country's US HIV aid is spent (ART, PMTCT, HIV prevention, testing and care, OVC) at the model's unit costs. No cut fraction, no `mou_us_ref` leverage, no 0% floor: money above the Original MOU amount restores what the Original MOU cut and then raises coverage above the 2021-23 baseline, capped at 100% of need. The 2027 total is the same in every ceiling (US$2,469,904,824; the ceilings differ from it by at most US$0.01). Conversions come from the model's marginal run (`marginal_2027_v2.csv`) divided by the HIV share of each country's US direct aid; HIV deaths and infections are a linear approximation at central parameters.

1. Totals, 15 countries: change vs the Original MOU

Positive = better than the Original MOU (more people on the service, more infections or deaths averted).

Scenario	HIV money moved (US$M)	ART patients 2027	ART coverage pp	PMTCT mothers 2027	PMTCT coverage pp	HIV infections averted 2026-30	HIV deaths averted 2026-30
Low Ceiling	44.5	-430	-0.00	+1,666	+0.43	+5,246	+245
Base Ceiling	89.1	-3,102	-0.03	+3,089	+0.80	+11,224	+363
Higher Ceiling	133.6	-34,906	-0.31	+3,563	+0.92	+10,271	-916

For scale, under the Original MOU the 15 countries have 118,793 HIV deaths and 527,414 new HIV infections over 2026-2030, and 455,448 people losing ART in 2027.

2. By country

Low Ceiling

Country	Δ HIV US$M	Δ ART patients	ART coverage 2027: baseline / Original → new	Δ PMTCT mothers	PMTCT coverage: Original → new	HIV infections averted	HIV deaths averted
Burundi	+1.53	+2,153	89.4% / 81.3% → 83.9%	+57	43.9% → 45.6%	+310	+115
Cameroon	-3.30	-9,328	73.0% / 66.3% → 64.6%	-79	67.3% → 66.8%	-1,414	-487
Kenya	-19.41	-31,444	91.2% / 90.0% → 87.9%	-1,190	89.6% → 86.8%	-2,993	-1,648
Mozambique	+43.02	+99,038	83.7% / 80.8% → 84.8%	+3,262	87.8% → 90.5%	+18,817	+5,413
Nigeria	-21.83	-60,849	78.1% / 75.0% → 72.0%	-385	60.4% → 59.5%	-9,474	-3,148

Base Ceiling

Country	Δ HIV US$M	Δ ART patients	ART coverage 2027: baseline / Original → new	Δ PMTCT mothers	PMTCT coverage: Original → new	HIV infections averted	HIV deaths averted
Burundi	+1.53	+2,153	89.4% / 81.3% → 83.9%	+57	43.9% → 45.6%	+310	+115
Cameroon	-6.60	-18,657	73.0% / 66.3% → 62.9%	-158	67.3% → 66.4%	-2,827	-975
Kenya	-38.83	-62,889	91.2% / 90.0% → 85.8%	-2,379	89.6% → 83.9%	-5,986	-3,295
Lesotho	+2.78	+5,694	96.1% / 79.5% → 81.8%	+124	77.1% → 79.2%	+2,297	+320
Mozambique	+77.31	+177,978	83.7% / 80.8% → 88.0%	+5,862	87.8% → 92.6%	+33,815	+9,727
Nigeria	-43.67	-121,697	78.1% / 75.0% → 68.9%	-769	60.4% → 58.7%	-18,948	-6,297
Sierra Leone	+3.63	+8,275	83.9% / 79.6% → 88.6%	+0	116.6% → 116.6%	+1,445	+421
Uganda	+3.85	+6,041	85.2% / 82.4% → 82.8%	+353	96.9% → 97.4%	+1,119	+346

Higher Ceiling

Country	Δ HIV US$M	Δ ART patients	ART coverage 2027: baseline / Original → new	Δ PMTCT mothers	PMTCT coverage: Original → new	HIV infections averted	HIV deaths averted
Burundi	+1.53	+2,153	89.4% / 81.3% → 83.9%	+57	43.9% → 45.6%	+310	+115
Cameroon	-9.89	-27,985	73.0% / 66.3% → 61.2%	-237	67.3% → 65.9%	-4,241	-1,462
Kenya	-58.24	-94,333	91.2% / 90.0% → 83.7%	-3,569	89.6% → 81.1%	-8,980	-4,943

Country	Δ HIV US$M	Δ ART patients	ART coverage 2027: baseline / Original → new	Δ PMTCT mothers	PMTCT coverage: Original → new	HIV infections averted	HIV deaths averted
Lesotho	+2.78	+5,694	96.1% / 79.5% → 81.8%	+124	77.1% → 79.2%	+2,297	+320
Mozambique	+77.31	+177,978	83.7% / 80.8% → 88.0%	+5,862	87.8% → 92.6%	+33,815	+9,727
Nigeria	-65.50	-182,546	78.1% / 75.0% → 65.8%	-1,154	60.4% → 57.8%	-28,423	-9,445
Sierra Leone	+3.63	+8,275	83.9% / 79.6% → 88.6%	+0	116.6% → 116.6%	+1,445	+421
Uganda	+48.40	+75,859	85.2% / 82.4% → 87.3%	+2,480	96.9% → 100.2% (capped)	+14,047	+4,350

Countries whose ART coverage in 2027 ends above its 2021-23 baseline: Mozambique (Low Ceiling: 83.7% → 84.8%), Mozambique (Base Ceiling: 83.7% → 88.0%), Mozambique (Higher Ceiling: 83.7% → 88.0%), Sierra Leone (Base Ceiling: 83.9% → 88.6%), Sierra Leone (Higher Ceiling: 83.9% → 88.6%), Uganda (Higher Ceiling: 85.2% → 87.3%).

Services where the gain is limited by 100% of need: Higher Ceiling Uganda PMTCT.

3. What US$1M of HIV aid buys in each country (Original MOU level)

ART and PMTCT: people on the service in 2027. Infections and deaths: averted over 2026-2030. Money split: share of the HIV dollar landing on each line in the model (the rest of each line is programme management and systems money the model spreads across lines).

Country	ART patients	ART cost US$/patient-year	PMTCT mothers	HIV infections averted	HIV deaths averted	ART share	Testing & care share	Prevention share	PMTCT share	OVC share	HIV share of US direct aid	Role in the ceilings
Cameroon	2,829	216	24	429	147.8	63%	18%	5%	1%	4%	78%	gives
Nigeria	2,787	208	18	434	144.2	53%	31%	5%	1%	4%	73%	gives
Malawi	2,399	199	30	314	124.6	52%	9%	16%	1%	2%	89%	unchanged
Mozambique	2,302	227	76	437	125.8	50%	24%	15%	2%	1%	82%	receives
Sierra Leone	2,281	191	0	398	116.1	40%	22%	15%	0%	0%	37%	receives
Eswatini	2,133	191	31	418	113.0	39%	17%	15%	1%	5%	95%	unchanged
Lesotho	2,052	191	45	828	115.3	31%	29%	15%	2%	4%	99%	receives
Rwanda	1,657	208	23	417	167.8	47%	9%	16%	1%	4%	78%	unchanged
Kenya	1,620	308	61	154	84.9	44%	29%	11%	2%	3%	87%	gives
Uganda	1,567	314	92	290	89.9	48%	21%	15%	3%	3%	84%	receives
Botswana	1,524	208	11	1,227	154.1	28%	18%	26%	0%	2%	99%	unchanged
Ethiopia	1,451	314	6	190	142.2	39%	29%	6%	0%	6%	67%	unchanged
Burundi	1,410	299	37	203	75.4	34%	32%	4%	3%	5%	68%	receives
Liberia	869	400	10	126	45.9	22%	37%	4%	1%	0%	39%	unchanged
Madagascar	9	400	0	32	1.6	0%	0%	20%	0%	1%	3%	unchanged

Madagascar's US aid is 3% HIV, so its HIV conversion rests on very little money and is not meaningful; it is unchanged in every ceiling.